習題 4
總體經濟學原理與實習

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Question 1

1-(a)

文中所指的利率爲 0.13%

$$(1.0013)^x = 2 \implies x \approx 534 \text{ (years)}$$

1-(b)

$$(1.008)^x = 2 \implies x \approx 87 \text{ (years)}$$

1-(c)

1. 儲蓄更少，投入更多在當期的消費支出上
2. 儲蓄更多，以補足所獲得的利息
3. 轉向風險更高的投資（如股票等）

Question 2

2-(a)

第一期之預算限制式爲

$$b_0(1 + R_0) + m_0 + p_1y_1 = p_1c_1 + b_1 + m_1$$

其中 $b_0 = 150$萬, $m_0 = 2$萬, $R_0 = 2\%$, $p_1y_1 = 100$萬, $p_1c_1 = 60$萬, $m_1 = 3$萬

2-(b)

$$\text{名目儲蓄} = b_0R_0 + p_1y_1 - p_1c_1 = 3 + 100 - 60 = 43(\text{萬元})$$

2-(c)

$$b_1 = 150(1.02) + 2 + 100 - 63 = 192(\text{萬元})$$

2-(d)

$$p_1y_1 - p_1c_1 = (b_1 + m_1) - (b_0 + m_0)$$
$$100 - 60 = (b_1 + 3) - (0 + 2)$$
$$b_1 = 39(\text{萬元})$$

2-(e)

因爲儲蓄爲資產的變動，固定資產投資前後價值不變，因此名目儲蓄仍爲 43 萬元。

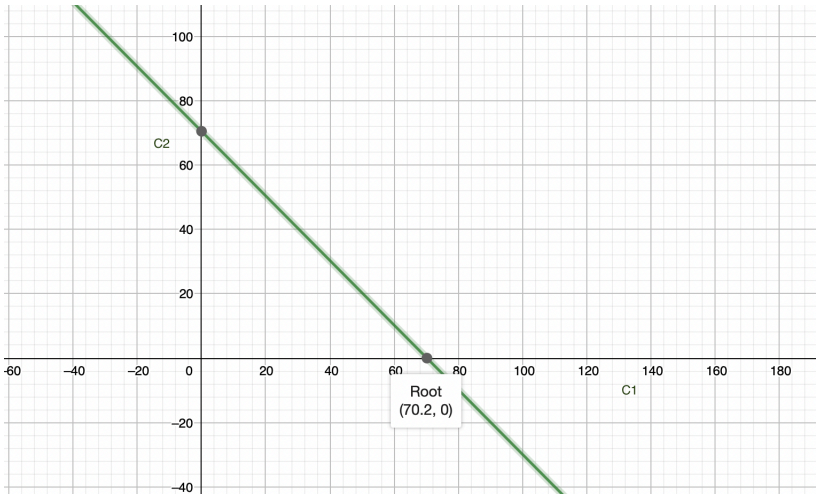
Question 3

3-(a)

$$b_0(1 + R_0) + p_1y_1 + \frac{p_2y_2}{1 + R_1} = p_1c_1 + \frac{p_2c_2}{1 + R_1}$$
$$40(1.005) + 30 + 0 = c_1 + \frac{c_2}{1.005}$$

與橫軸焦點的截距：$c_2 = 0 \implies$ 所求

$$40(1.005) + 30 = c_1 = 70.2 = \text{與橫軸之截距}$$



3-(b)

第一期儲蓄 $= 40(0.002) + 30 - 40 = -9.2$
$b_1 - b_0 = -9.2 \implies b_1 = 30.8$萬元

since $b_2 = 0$,

$$b_2 - b_1 = (b_1(0.002) - c_2)$$

$$0 - 30.8 = 0.0616 - c_2$$

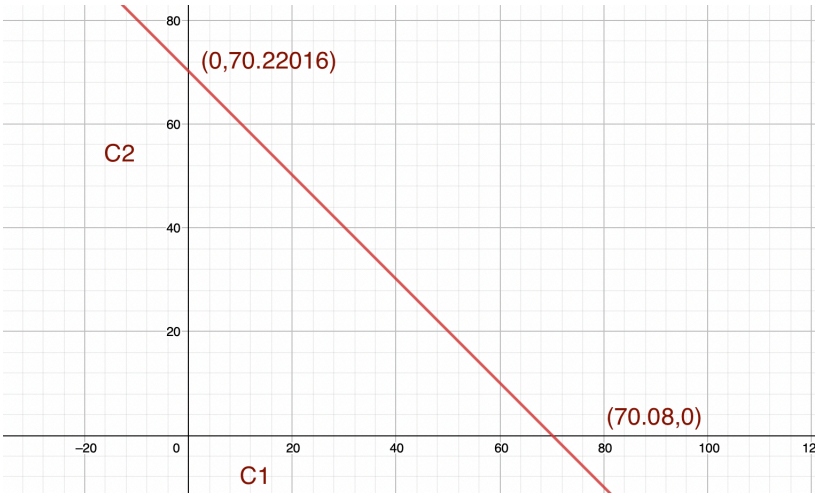
$$c_2 = 30.8616(\text{萬元})$$

3-(c)

新的預算限制線

$$b_0(1 + R_0) + p_1 y_1 + \frac{p_2 y_2}{1 + R_1} = p_1 c_1 + \frac{p_2 c_2}{1 + R_1}$$

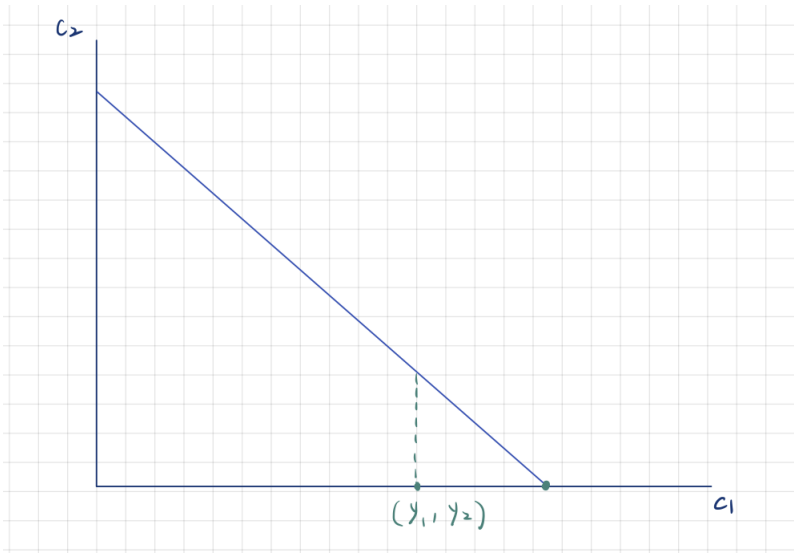
$$40(1.002) + 30 + 0 = c_1 + \frac{c_2}{1.002}$$



利率下降後，所有 (c_1, c_2) 的消費組合有下降的現象，代表購買力下降

Question 4

4-(a)

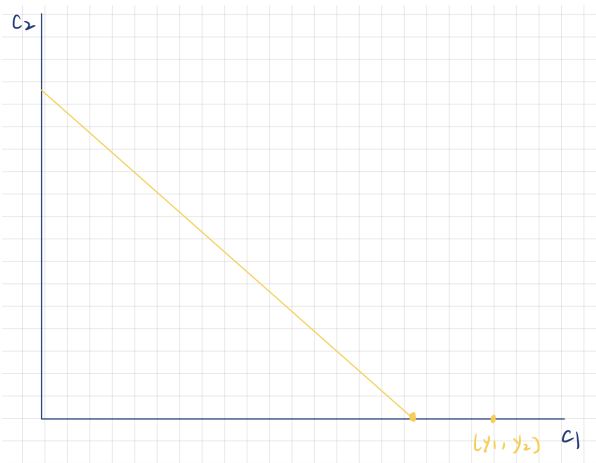


4-(b)

$$c_2 = \frac{b_0(1+R_0)}{p_1}(1+r_1)$$

4-(c)

$$b_0 < 0 \implies \text{x 截距} < y_1$$



由圖，顯然可知 $c_1 = y_1$ 的選擇是不可能的

Question 5

5-(a)

Suppose that $b_0 = b_2 = 0$, 第一期的預算限制式：

$$p_1 y_1 = p_1 c_1 + b_1$$

第二期的預算限制式：

$$b_1 + p_2 y_2 = p_2 c_2$$

兩式相加：

$$p_1 y_1 + p_2 y_2 = p_1 c_1 + p_2 c_2$$

5-(b)

Suppose that $b_0 = b_2 = 0$, 第一期的預算限制式：

$$p_1 y_1 = p_1 c_1 + b_1$$

第二期的預算限制式：

$$b_1 + \frac{p_2 y_2}{1 + R_1} = \frac{p_2 c_2}{1 + R_1}$$

兩式相加：

$$p_1 y_1 + \frac{p_2 y_2}{1 + R_1} = p_1 c_1 + \frac{p_2 c_2}{1 + R_1}$$

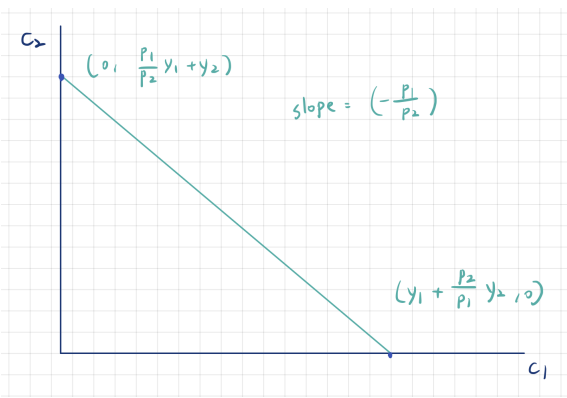


Figure 1: 5(a) 之作圖

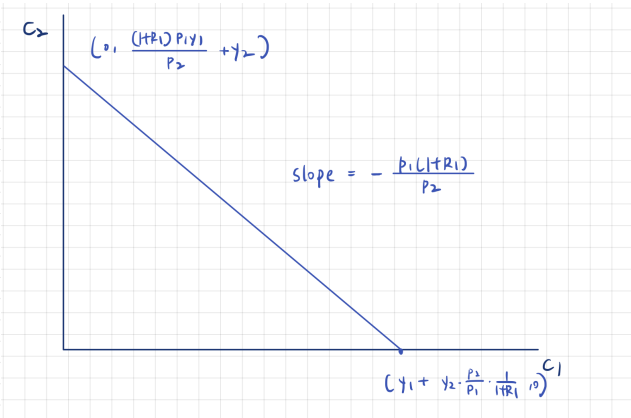


Figure 2: 5(b) 之作圖